Constraint Lifecycle Taxonomy

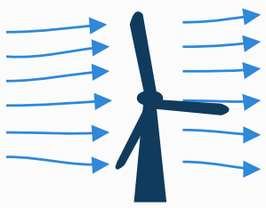
Five orthogonal dimensions for auditing physics-constrained forecasting studies

D1 Parameter Provenance

Where does the constraint originate?

Intrinsic

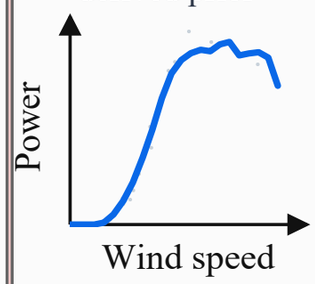
Governing laws / physical limits



$C_p \leq 16/27$

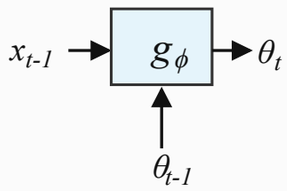
Historical-static

Offline SCADA-derived prior



Learned-online

Adaptive learned parameters



online update

Unreported

Calibration omitted

Not specified

D2 Application Stage

Where does physics enter the forecasting system?

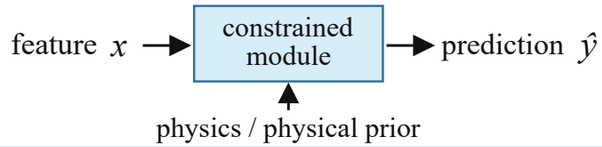
Training loss

Physics term in objective function

$$\mathscr{L} = \mathscr{L}_{task} + \lambda \mathscr{L}_{phys}$$

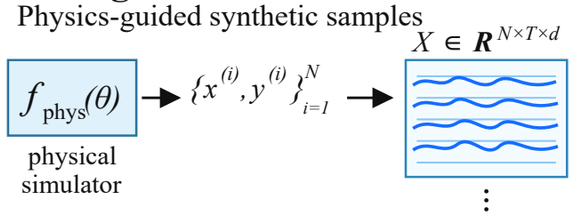
Architecture

Physics-constrained layer / graph



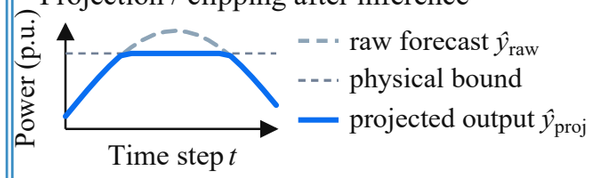
Data generation

Physics-guided synthetic samples



Post-hoc correction

Projection / clipping after inference

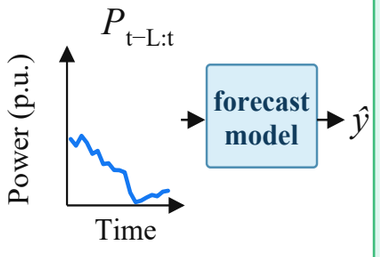


D3 Inference Telemetry Requirements

What external information is required at deployment?

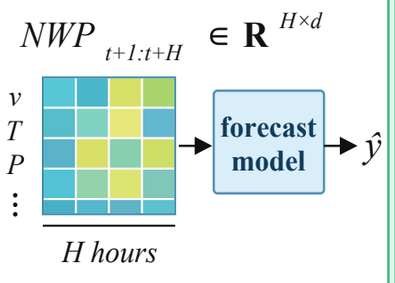
None

Historical target SCADA



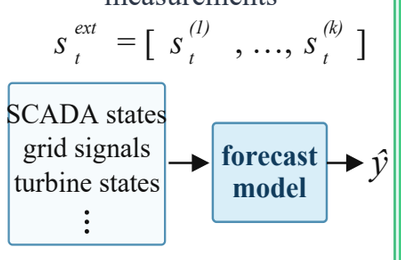
Forecast input

Future NWP



Contemporaneous state

Synchronized external measurements



D4 Implementation Fidelity

Can the published formulation be verified in executable code?

Code availability

Public

Source code available

```
loss.py
return ℒ_phys
```

Not found

No accessible source code

Repository omitted

(Code not public)

Equation–code parity

If code is public, evaluate equation–code parity

Published equation $\mathscr{L}_{pub}(\theta)$ ◆ **Source code** $\mathscr{L}_{exec}(\theta)$

Aligned

Executable implementation matches formulation

$\mathscr{L}_{pub}(\theta) = \mathscr{L}_{exec}(\theta)$

Mismatch

Discrepancy between formulation and code

$\mathscr{L}_{pub}(\theta) \neq \mathscr{L}_{exec}(\theta)$

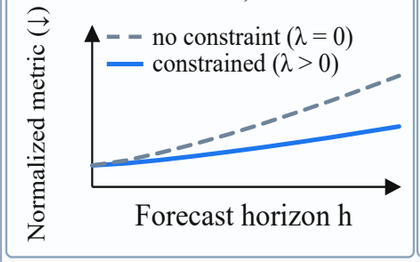
D5 Evaluation Validity & Harm Reporting

Does the evaluation isolate the physical prior and expose failure regimes?

Ablation semantics

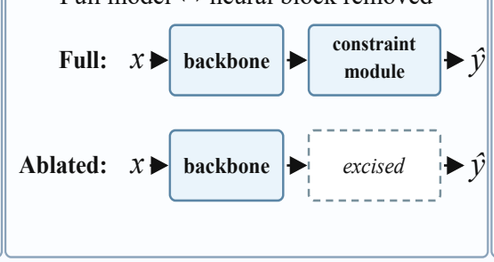
Constraint on/off

Identical backbone; $\lambda = 0 \leftrightarrow \lambda > 0$



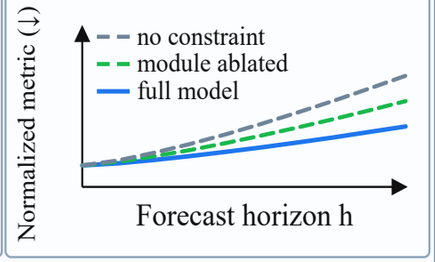
Module ablation

Full model ↔ neural block removed



Both

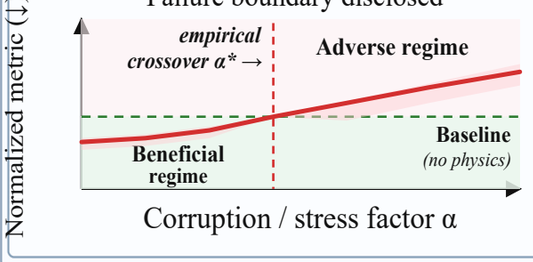
Constraint on/off & module ablation



Failure / harm reporting

Reported

Failure boundary disclosed



Not reported

Failure boundary omitted

